



Toward Agentic AI: NeMo Agent Toolkit

Iven Fu | Solutions Architect, NVAITC



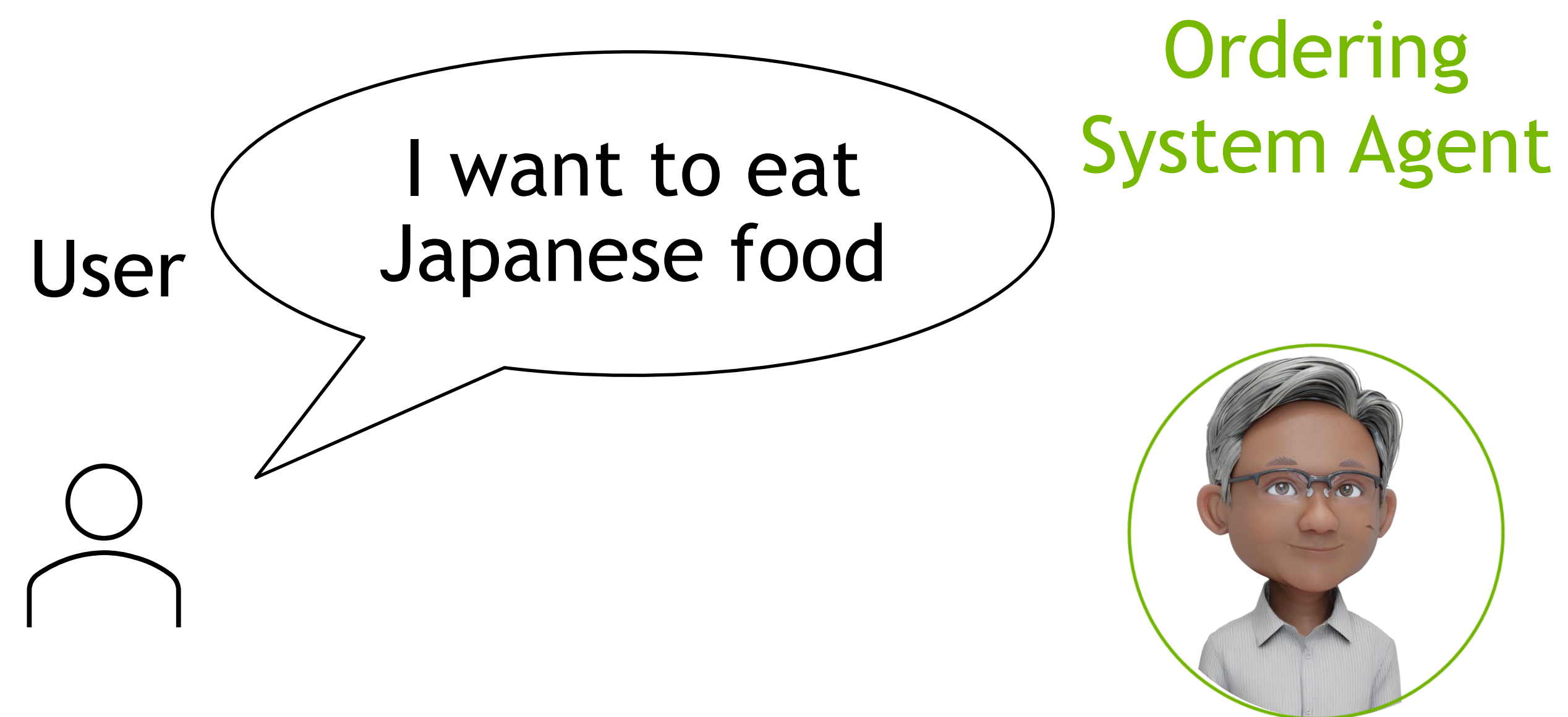
Agentic Models

Agentic Models

- Agentic AI is humanity's ideal for artificial intelligence
 - Achieve complex goal with minimum human intervention
 - It will interact with the **environment**, and do **planning**, use **tools**, **make decisions**, or **adaptively change its actions** according to environment observation

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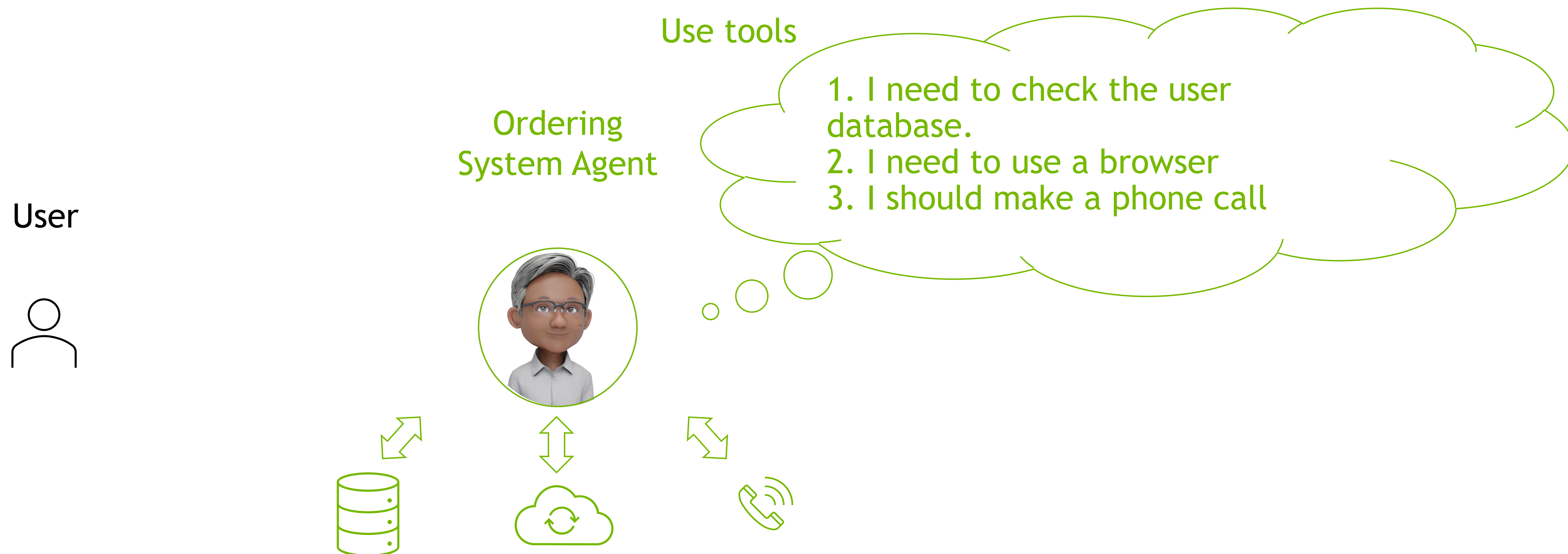
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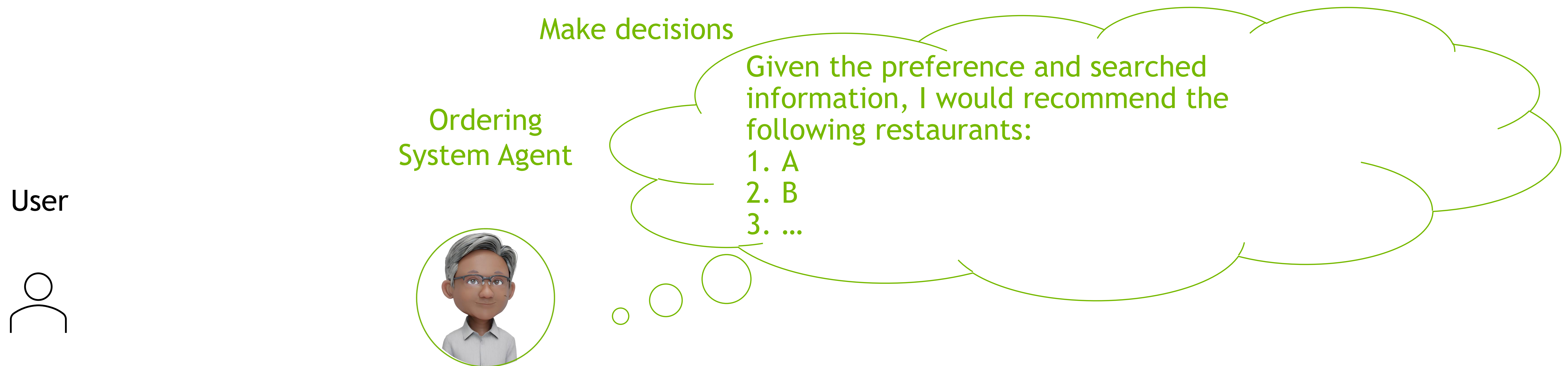
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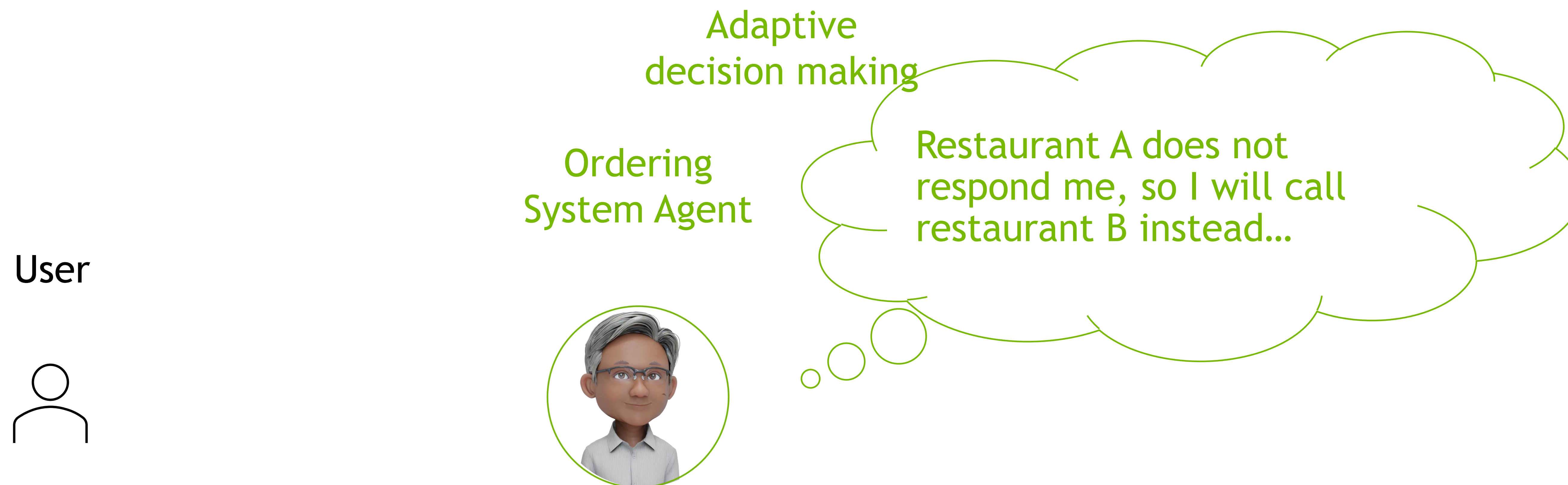
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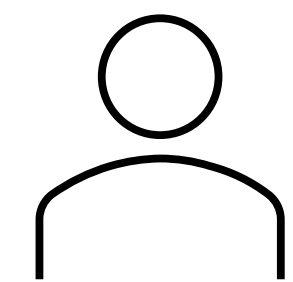
Agentic Models Orchestration

- Coordinates **multiple** agents to work together toward a goal
- Handles task assignment, **sequencing**, and **collaboration**
- Enables **complex** workflows (e.g., research, planning, execution)

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User



I want to eat
Japanese food

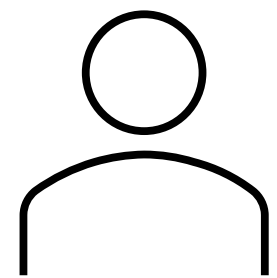
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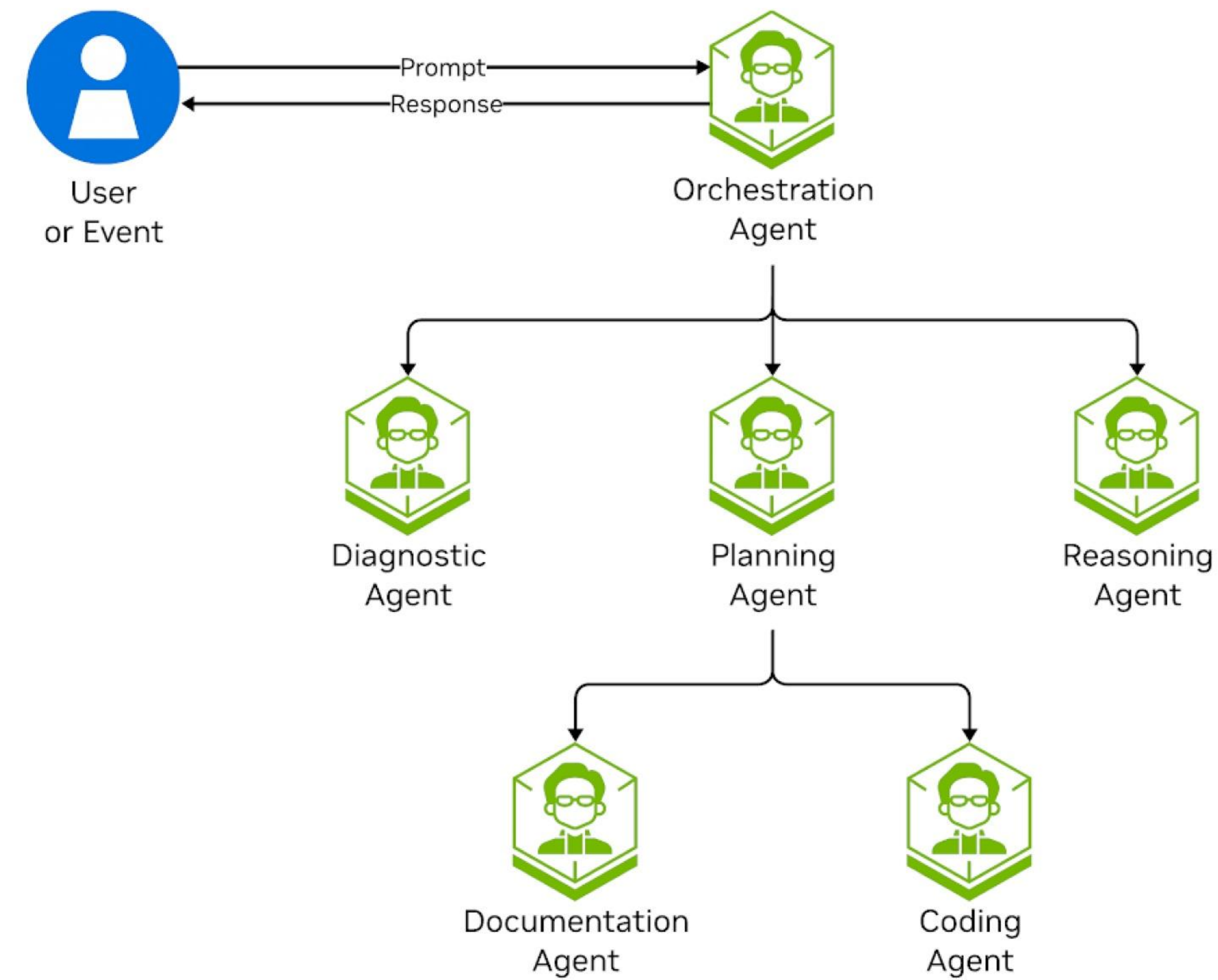


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Tools



Bug Database



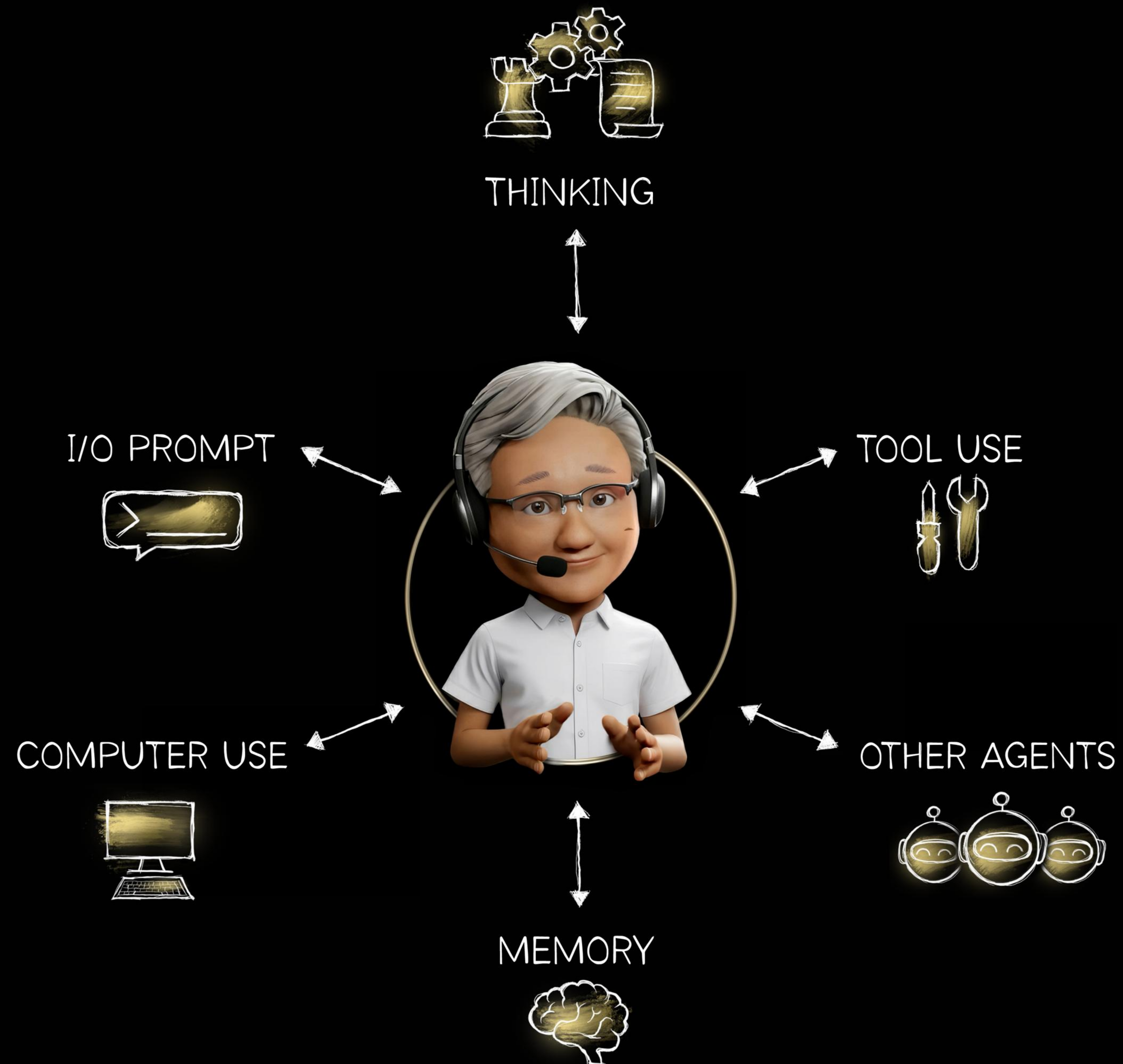
Documentation



Code Repo



RAG





NeMo Agent Toolkit

NeMo Agent Toolkit Reduces Agentic System Complexity

Flexible integration, configuration, monitoring, and optimization



Framework Agnostic

Multi-framework support: LangChain, LangGraph, CrewAI, Semantic Kernel, Google ADK



Evaluation System

Customizable evaluation framework for agent workflow components



Observability Tracing

End-to-end tracing across frameworks and agent invocations



Profiler

Performance analysis and runtime bottleneck identification



Workflow Optimization

Agent Hyperparameter Optimizer, NVIDIA Dynamo integration for accelerated performance, Reinforcement learning



Safety and Security

Evaluate risk profile, find vulnerabilities, red-team, mitigate risks, apply defense